Practical Assignment (problem solving questions)

Level: Year 2 – Medical Students

| | N | P | |
|---|---|---|---|
| d | 5 | 10 | 15 |
| nd | 45 | 40 | 85 |
| | 50 | 50 | 100 |

1- 100 people are tested for the disease. 15 people have the disease; 85 people are not diseased. 50 patients were positive for the test. Among 85 individual found not to be diseased according to the gold standard, 40 individual were found to be positive for the test, calculate Prevalence of Disease, Sensitivity of the test & Specificity of the test.

sensitivity $= \frac{TP}{\text{diseased}} = \frac{50-40}{15} = \frac{10}{15} = \frac{2}{3} = 66.67\%$

specificity $= \frac{TN}{\text{Non diseased}} = \frac{45}{85} = 52.9\%$

prevalence $= \frac{\text{diseased}}{\text{total}} = \frac{15}{100} = 15\%$

2- Many of the students at the boarding school, including 6 just coming down with varicella, went home during the Thanksgiving break. About 2 weeks later, 4 siblings of these 6 students (out of a total of 10 siblings) developed varicella. Calculate the secondary attack rate among siblings

secondary attack rate $= \frac{\text{New cases among contact}}{\text{total cases at risk}} \times 100$

$\frac{4}{10} \times 100 = 40\%$

3- A new screening test was evaluated for Disease A among 500 individuals using a gold standard.

- According to the gold standard, 80 individuals had Disease A.
- The screening test was positive in 60 diseased individuals.
- Among the 420 non-diseased individuals, 90 tested positive.

| | P | N | |
|---|---|---|---|
| d | 60 | 20 | 80 |
| Nd | 90 | 330 | 420 |
| | 150 | 350 | 500 |

1. Calculate the prevalence of Disease A. $= \frac{\text{existing cases}}{\text{total pop}} = 80$
2. Construct a 2 × 2 table.
3. Calculate:

| | diseased | Not | total |
|---|---|---|---|
| +Ve | 60 | 90 | 150 |
| -Ve | 20 | 330 | 350 |
| total | 80 | 420 | 500 |

Sensitivity, Specificity, Positive Predictive Value (PPV), Negative Predictive Value (NPV)

sensitivity = $\frac{TP}{disease} = \frac{60}{80} = 75\%$

specificity = $\frac{TN}{not\ disease} = \frac{330}{420} = 78.6\%$

PPV = $\frac{TP}{total\ P} = \frac{60}{150} = 40\%$

NPV = $\frac{TN}{total\ N} = \frac{330}{350} = 94.3\%$

4- In a community screening program:

- Disease prevalence = 8%
- Total population screened = 2,500
- The screening test correctly identified 160 true cases.
- 2,070 individuals were correctly identified as disease-free.

Questions:

1. Calculate the number of diseased and non-diseased individuals.
2. Complete the 2 × 2 table.
3. Calculate sensitivity, specificity, PPV, and NPV.
4. Based on PPV and NPV, state whether the test is more suitable for screening or diagnosis.

disease prevalence = $\frac{total\ cases}{total\ pop} = 8\% = \frac{total\ cases}{2500}$

total cases = 200

non diseased = total pop − diseased = 2500 − 200 = 2300

| | diseased | not D | total |
|---|---|---|---|
| +Ve | 160 | 230 | 340 |
| -Ve | 40 | 2070 | 2110 |
| total | 200 | 2300 | 2500 |

sensivity = $\frac{TP}{disease} = \frac{160}{200} = 80\%$

specificity = $\frac{TN}{not\ D} = \frac{2070}{2300} = 90\%$

PPV = $\frac{TP}{total\ P} = \frac{160}{340} = 41\%$

NPV = $\frac{TN}{total\ N} = \frac{2070}{2110} = 98.1\%$

5- In City B, the following data were recorded in 2023:

- Total population: 90,000
- Women aged 15–49 years: 24,000
- Married women aged 15–49 years: 18,000
- Total live births: 2,700

| Age group (years) | Live births | Women |
|---|---|---|
| 15–19 | 135 | 5,000 |
| 20–24 | 720 | 6,000 |
| 25–29 | 810 | 5,000 |
| 30–34 | 540 | 4,000 |
| 35–39 | 315 | 2,500 |
| 40–44 | 135 | 1,200 |
| 45–49 | 45 | 300 |

1. Calculate the Crude Birth Rate (CBR).
2. Calculate the General Fertility Rate (GFR).
3. Calculate ASFRs.
4. Calculate the Total Fertility Rate (TFR).

$TFR = \frac{ASFR \times 5}{1000} = \frac{(27+120+162+135+126+112.5+15)+5}{1000} = 4.16$ children per woman

1) $\frac{\text{live birth}}{\text{total pop}} \times 1000 = \frac{2700}{90000} \times 1000 = 30$ per 1000

2) $\frac{\text{live birth}}{\text{women 15-49}} \times 1000 = \frac{2700}{24000} \times 1000 = 112.5$ per 1000

3) $\frac{\text{Live birth in group}}{\text{women in same group}} \times 1000 = \frac{540}{4000} \times 1000 = 135$ per 1000

6- A health survey in a town with a population of 40,000 revealed:

- Existing cases of hypertension at start of year: 3,200
- New hypertension cases during the year: 400
- Population at risk: 36,800

- Total episodes of acute respiratory infection (ARI): 2,400
- Number of persons affected by ARI: 2,000
- Total days lost due to ARI: 4,800
- Population screened for anemia: 3,000
- Anemia cases detected: 450

1. Calculate point prevalence of hypertension.
2. Calculate period prevalence of hypertension.
3. Calculate incidence rate of hypertension.
4. Calculate attack rate of ARI.
5. Calculate prevalence of anemia.

1- $\frac{3200}{40000} \times 100 = 8\%$

3- $\frac{400}{36800} \times 1000 = 10.87\%$

2- $\frac{3200 + 400}{40000} \times 100 = 9\%$

4- $\frac{2000}{40000} \times 100 = 5\%$

5- $\frac{450}{3000} \times 100 = 15\%$

**A depression screening test is compared blindly with a gold standard in 200 psychiatric outpatients. Of 50 true depressed patients, 35 tested positive; of 150 non-depressed individuals, 30 tested positive.** calculate the following:

1. Sensitivity
2. Specificity
3. Positive Predictive Value (PPV)
4. Negative Predictive Value (NPV)
5. Prevalence rate of depression

1) $\frac{TP}{Disease} = \frac{35}{50} = 70\%$

3) $\frac{TP}{total\ P} = \frac{35}{65} = 53.8\%$

2) $\frac{TN}{Non\ D} = \frac{120}{150} = 80\%$

4) $\frac{TN}{total\ N} = \frac{120}{140} = 88.9\%$

| | P | ND | |
|---|---|---|---|
| + | 35 | 30 | 65 |
| - | 15 | 120 | 140 |
| | 50 | 150 | 200 |

A sudden increase in cases of acute gastroenteritis is reported among students in a boarding school over 3 days. Many students present with vomiting, diarrhea, and abdominal cramps, describe the **steps you would follow to investigate this epidemic**.

1- verify the diagnosis
2- establish the existence of outbreak
3- Define & identify cases
4- perform descriptive epidemiology
5- develop hypotheses
6- Evaluate hypotheses
7- environmental investigation
8- implement control & prevention
9- communicate Findings

In a table form **Compare between a common source outbreak and a propagated epidemic** in terms

1. Source of infection
2. Mode of transmission
3. Epidemic curve pattern
4. Incubation period distribution
5. Duration of outbreak
6. Main control measures

| | Common source outbreak | Propagated epidemic |
|---|---|---|
| source | single common source | Person to Person spread |
| transmission | indirect (vechile-born) | Direct or vector |
| epidemi curve | sharp rise, Rapid decline | Gradual rise multiple peaks |
| Duration | short | prolonged |
| incubation | one | multiple |
| main control | eliminate the source | isolation, Vaccination |

During a wedding party, several attendees developed acute gastrointestinal illness characterized by vomiting and diarrhea within 12 hours of the event. A cohort study was conducted to identify the food item responsible for the outbreak.

The following data were collected:

| Food Item | Ate Food & Became Ill | Ate Food & Not Ill | Did Not Eat & Became Ill | Did Not Eat & Not Ill |
|---|---|---|---|---|
| Chicken | 40 | 20 | 10 | 30 |
| Salad | 35 | 25 | 15 | 25 |
| Cake | 50 | 10 | 5 | 40 |

**Questions**

1. Calculate the **Relative Risk (RR)** for each food item.
2. Interpret the RR values.
3. Identify the **most likely food item responsible** for the outbreak.
4. Justify your answer based on epidemiological principles.

$$RR = \frac{\text{incidence in exposed}}{\text{incidence in unexposed}}$$

RR chicken → 2.67
RR Salade → 1.55
RR Cak → 7.5

chiken (2.67) who ate it were 2.67 more like to become ill than who didn't

3. Cake

4. 1. trength of association → the higher RR to stronger association
2. Attake rate difference → incidence of who are ake higher that who didn't
3. statistical significance → RR for 7.5 indicate that cake is primary cause of the out break